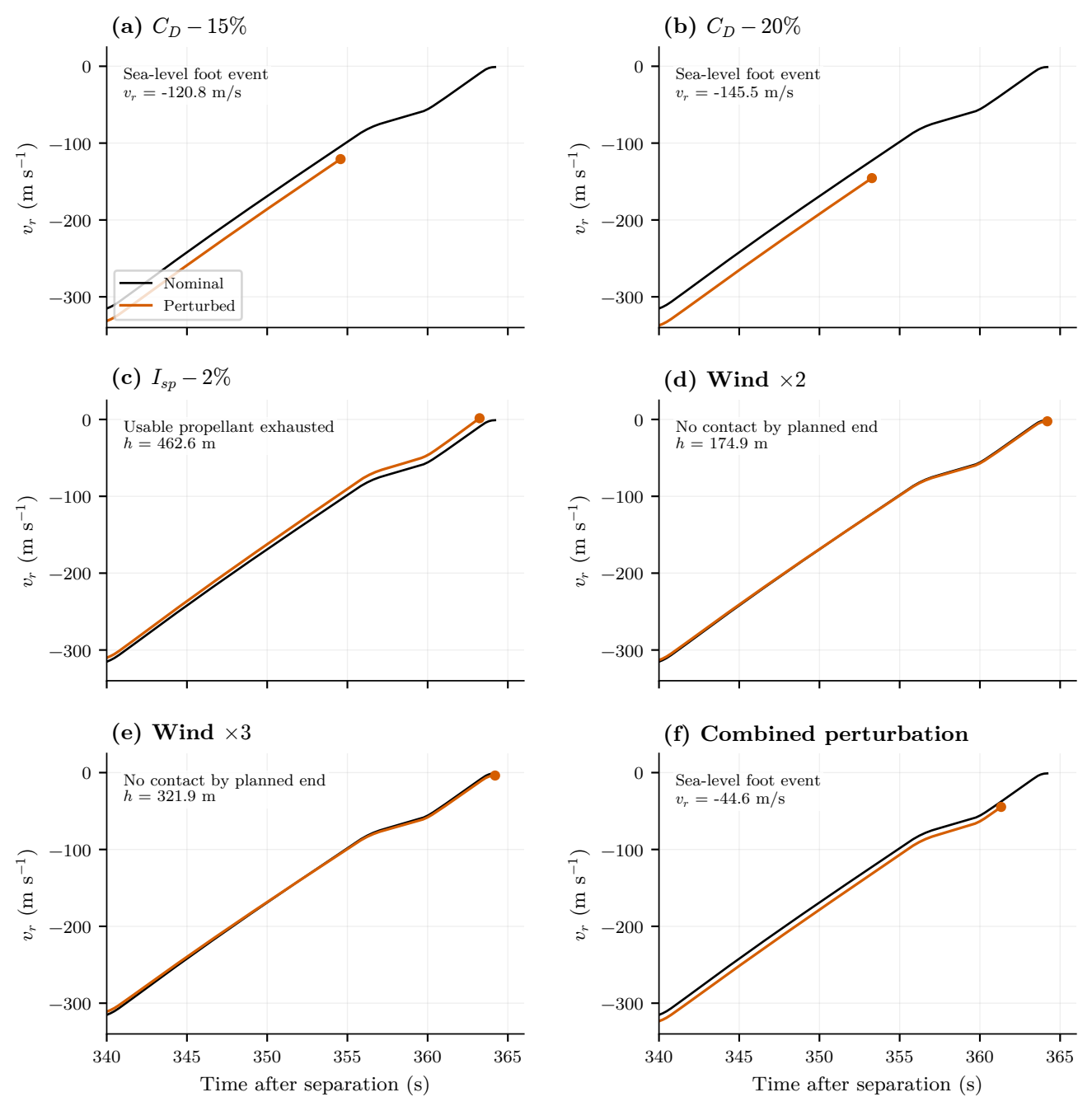
(a) $C_D - 15\%$
Sea-level foot event
v_r = -120.8 m/s
Nominal
Perturbed
(b) $C_D - 20\%$
Sea-level foot event
v_r = -145.5 m/s
(c) $I_{sp} - 2\%$
Usable propellant exhausted
h = 462.6 m
(d) Wind ×2
No contact by planned end
h = 174.9 m
(e) Wind ×3
No contact by planned end
h = 321.9 m
(f) Combined perturbation
Sea-level foot event
v_r = -44.6 m/s
v_r (m s^{-1})
0
−100
−200
−300
340
345
350
355
360
365
Time after separation (s)